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(1) Research Program/Project/Study Title: Development of Solar Powered Vendo E-charging Station	
Proponent Name:	Stephen T. Bascos
Academic Rank:	Instructor I
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education/Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/Computer/Engineering/Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation			
		Municipality/City:	Roxas City
		Province:	Capiz
		Region:	VI
(5) Classification		(6) Mode of Implementation	
Research:	Development:	☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
☒ Basic ☐ Applied	☐ Pilot Testing ☐ Tech. Promotion/Commercialization		
(7) Sector/Commodity		(8) Discipline/Field of Specialization	
Electrical/Electronics Sector		Electronics Technology	

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(9) Rationale(State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

- E-vehicles, mobile phones and other smart devices keep on running all the time anywhere and everywhere, draining its battery. Recharging them need certain time and suitable place.
- Sudden shutdown of those gadget and devices due to lack of charge, creates huge disappointment for people who is in rush to their workplace, market, school, college, office, train and bus station and so on.
- It would be great if we could facilitate these people with a charging station that can convenient to their charging needs with the help of renewable energy harvester which they can avail whenever on the go.
- Most of these can be applicable for public use but some can be intended for commercial purposes.
- In case of calamities and other natural disaster occurs, this device can be used as an emergency charging station in the event of prolonged shortage and outage of electrical power energy sources

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study aims to develop a **Solar powered vendo e-charging station**. Specifically, it seeks to find out the following:

- Determine the device charging capacity in terms of current state and discharging capacity in terms of power utilization in charging per hour (a. e-bike b. laptops c. mobile phone and other gadgets)
- Find out the acceptability level of the device in terms of technical specifications, operating performance and safety.

(11) Review of Literature/Prior Art Search(Related researchers/activities that have been conducted. State-of – the-art of current technology / information from which the project proposal will take-off).

These are the prior arts and related literature that the study was based. The researcher gathered the selected articles that provided useful and further insights about the study.

Personal electronic charging station - LOEWEN JOANNE [US]

The personal electronic charging station is adapted for use in charging one or more personal data devices. The personal electronic charging station is adapted for use one or more electronic payment systems selected from the group consisting of credit card payment systems or a token based payment system. The personal electronic charging station is a switched charging station that distributes electric power to the personal data devices on a fee for service basis. Once the fee has been received through the electronic payment system, the personal electronic charging station will distribute electrical power for use in charging the one or more personal data devices. The personal electronic charging station comprises a cabinet, a charging circuit, and a control system.

Solar powered charging station for electric and plug-in hybrid vehicles - Paluszek Michael A [US]; et.al.

This invention is a solar-powered charging station for electric and hybrid vehicles. A vehicle parks at a space with the charging station and uses a credit card, debit card, cash, smart card or network connection to a database like EZ-Pass to pay for the space and the electricity. The station automatically charges the vehicle as long as it is connected to the station. The station automatically stops charging when the vehicle is fully charged. The customer only pays for the space and the electricity consumed. If the charging circuit is broken the customer must reinsert the smart card or credit card to restart charging. Sufficient funds are removed from the payment method on initiation of charging. Any money not used for charging is put back onto the smart card or account if the user reinserts it prior to leaving.

Solar Powered Charging Station - Hornsby David [US]

A self-contained charging station is provided which utilizes a solar panel to generate electrical current. The panel is integrated onto a utilitarian structure wherein one or more outlets for charging portable devices are provided.

Charging station for mobile device with solar panel - He Gang [US]; et.al.

A charging device configured to charge a mobile device through the solar cells integrated on the mobile device. The charging device converts wall power to light energy which can be absorbed by the solar cells and then converted to electricity for storage in the rechargeable battery of the mobile device. The charging device includes a

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light source configured to emit a light beam having a spectrum tuned to the spectral response of the solar cells. The charging device includes a proximity sensor for detecting the presence of a mobile device within the charging device housing and responsively signaling the activation of the light source. The charging device includes logic for wirelessly communicating with the mobile device as well as controlling the charging process in various stages and aspects. The light source may be LEDs that also serve to transmit light communication signals to the mobile device.

Electrified vehicle charging station configured to provide parking guidance to electrified vehicles - Lieberman Colin Turner [Us]

An electrified vehicle charging station according to an exemplary aspect of the present disclosure includes, among other things, a charger assembly including a plug connected to a terminal via a charger cable, and a transceiver configured to communicate information to an electrified vehicle which is useable to guide the electrified vehicle to an acceptable parking position. The acceptable parking position is a position in which the plug will reach a charging port of the electrified vehicle. A method is also disclosed.

Battery Management System for E-Vehicle Using Solar Power - Dr. J. Jaya, et.al.

An electric vehicle charging station, also called EV charging station, -electric recharging point, charging point, charge point, electronic charging station (ECS), and electric vehicle supply equipment (EVSE), is an element in an infrastructure that supplies electric energy for the recharging of plug-in electric vehicles—including electric cars, neighborhood electric vehicles and plug-in hybrids. For charging at home or work, some electric vehicles have converters on board that can plug into a standard electrical outlet or a high-capacity appliance outlet. Others either require or can use a charging station that provides electrical conversion, monitoring, or safety functionality. These stations are also needed when traveling, and many support faster charging at higher voltages and currents than are available from residential EVSEs. Public charging stations are typically on-street facilities provided by electric utility companies or located at retail shopping centers, restaurants and parking places, operated by a range of private companies.

(12) Scientific Basis/Theoretical Framework(Include concepts, theories and variables that indicate relevant relationships to be tested.

The solar charging stations convert clean free solar energy to electricity in areas where electrical outlets are unavailable. The solar panel converts sunlight into DC electricity to charge the backup storage battery which is attached to the charging station. This DC electricity is fed to the battery via a solar regulator which ensures the battery is charged properly and not damaged. It contained charger outlets can be used to recharge and power up devices.

(13) Methodology(Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

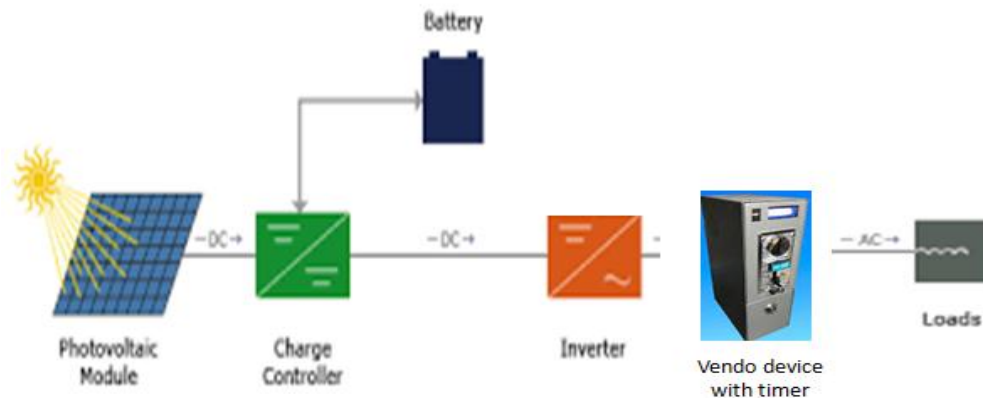
This study is a developmental research that involves developing or improving a certain product for the purpose of finding its effectiveness and usefulness for human assistance. The device is base from the principles of converting solar energy to electrical energy then fed in to the storage battery that can be utilize for charging low powered devices and gadgets. An innovation to be made for the device where it can charge various rechargeable units including e-bikes, laptop, mobile phones and other gadgets with a vendo machine for time controlled charging system.

Design Plan and Fabrication

The device is design for produce conventient and alternative power source where solar energy is converted into electrical energy and stored in a battery then produce a intended power output for charging e-bikes, laptop, mobile phones and other gadgets with the support of power electronic circuitry and vendo machine for time controlled charging system.

The device will includes the basic design of a solar PV system with some innovation on its charging operation. The researchers will illustrates the device on its conceptual design with the corresponding flow of operation in the charging station device.

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conceptual design of Solar Powered Vendo E-charging Station

Procedure in Fabricating the Solar Powered Vendo E-charging Station

- Plan the design and calculate the needed power output for the electrical load
- Gather solar power components including the vendo device
- Set up all the components according to their use in the pv system
- Fix and connect the solar panels to the pv system
- Test the product and calibrate the time setting for vendo device.

Instrumentation

With the Solar Powered Vendo E-charging Station as the convenient and portable charging station to determine the device charging and discharging capacity in terms of current state and power utilize in charging per hour of an e-bike, laptops, mobile phone and other gadgets

The observation and testing processes, the device charging and discharging capacity will be measure. For charging process, the device must be observe within charging state under sunlight within an hour then record the data. Likewise , the device will be monitor to determine how much current will draw within an hour during charging process on the e-bike, laptops, mobile phone and other gadgets.

On the power output utilize in charging various devices like e-bike, laptops, mobile phone and other gadgets within an hour, the voltage source shall be measure using measuring instrument then multiply to the observe current drawn. The evaluation process, the results for acceptability of the device in terms of technical specifications, operating performance and safety will be calculated to get the grand mean of acceptability using statistical tool.

Parameter for Analysis

The following parameters will be use to observe the operation of Solar Powered Vendo E-charging Station on its charging and discharging capacity in terms of current state and power delivered in charging per hour. For the current state, the amount of current fed in during charging and withdraw during discharging will be measure within a period of time using a formula of $I = q/t$, where I is the current, q is the charge and t is the time. The unit to be used in expressing the result will be Ah or Ampere-hour. For the power utilization during charging operation, the current drawn in the device will be multiply to the voltage output used by the clients rechargeable device.

For the evaluation process, through this way of questionnaire with a certain statement to be check by respondents according how he/she perceive and convince to answer the evaluation sheet in terms of technical specifications, operating performance and safety. In scoring of the variable or factors, the five (5) point's likert scale was used (Brooddifield and Moredock, 1957) the variable were scored, record and categorized as follows:

Scale	Mean	Verbal Interpretation
5	4.12 – 5.00	Very Acceptable
4	3.41 – 4.20	Acceptable
3	2.61 – 3.40	Moderately Acceptable
2	1.81 – 2.60	Less Acceptable
1	1.00 – 1.80	Not Acceptable

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Further, the quantitative data gathered through the observation and evaluation of Solar Powered Vendo E-charging Station will be ascertained with the use of the arithmetic mean as statistical tool.

(14) Major Activities/Work plan(Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1												Year 2											
	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	July	Aug
1. Planning of design and processes/Gathering of materials																								
2. Fabricating/testing/data gathering thru observation																								
3.Pilot testing/Evaluation/ Revision																								

(15) Expected Output(Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

This study will develop a device that can convenient to the public for their charging needs with the help of renewable energy harvester which they can avail whenever on the go.

The expected output of this study will surely provide an alternative sources of electricity or an emergency charging station in the event of charging needs and prolonged shortag/outrage of electrical power energy sources. The prototype of this device will be available for public use and for commercial purposes.

(15) Technology Transfer Activity Design(Indicate the specific products, activity design, processes or services to be transferred as extension activities)

This device will be offered free charging services to the trainees/clients of extension activities.

(16) Target Beneficiaries(Who the clientele are and what are the expected outcome/effects of the use of the projects output)

This device intended only for charging the batteries of the following clients;

- E-bike users (36VDC, 48 VDC and 64 VDC)
- Laptops, mobile phone and other gadgets owners/users (5VDC, 12VDC, 19VDC)

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Stephen T. Bascos	Study Leader	Electronics Technology

Percent time devoted to the Program/ Project/ Study: 90%

Program/Project Leader (in hours): 6 hours

Program/Project Members (in hours):

(18) Budgetary Requirement

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Materials	Quantity	Amount (PhP)
Solar Panel (32VDC out, 100W)	1 Panel	4, 500
PV Lithium Ion Battery (12VDC, 500mAh)	2 pc	10, 000 x 2
Solar PV Controller/Regulator (MPPT 100A)	1 unit	1,500
Power Inverter (2,500 watts)	1 unit	4, 500
Charging timer monitoring with Coin Slots	5 units	1, 000 x 5
Power Regulator (5VDC output, 2 USB ports)	1 unit	250
Charging connectors(12VDC Batteries)	1 set	150
Casing	Box	1000
Total		36, 900.00

(19) Literature Cited(include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

US10672218B1 Personal electronic charging station

<https://worldwide.espacenet.com/patent/search/family/070856315/publication/US10672218B1?q=US10672218>

US2022176839A1 ELECTRIFIED VEHICLE CHARGING STATION CONFIGURED TO PROVIDE PARKING GUIDANCE TO ELECTRIFIED VEHICLES

<https://worldwide.espacenet.com/patent/search/family/081655246/publication/US2022176839A1?q=US2022176839A1>

US2016204650A1 SOLAR POWERED CHARGING STATION

<https://worldwide.espacenet.com/patent/search/family/056368218/publication/US2016204650A1?q=US2016204650A1>

US9853489B2 Charging station for mobile device with solar panel

<https://worldwide.espacenet.com/patent/search/family/055315770/publication/US9853489B2?q=US9853489B2>

US2012313568A1 Solar Powered Charging Station For Electric And Plug-In Hybrid Vehicles

<https://worldwide.espacenet.com/patent/search/family/047292613/publication/US2012313568A1?q=US20120313568A1>

IN202241024394 - Battery Management System For E-Vehicle Using Solar Power

<https://Patentscope.Wipo.Int/Search/En/Detail.Jsf?Docid=In361380865& Cid=P20-L8ars8-67068-1>

<https://solenergy.com.ph/solar-pv-system-for-ev-charging-stations/>

<https://www.diva-portal.org/smash/get/diva2:1115565/FULLTEXT01.pdf>

<https://www.pveducation.org/pvcdrom/battery-characteristics/battery-charging-and-discharging-parameters#:~:text=In%20this%20case%2C%20the%20discharge,%2F20%20h%20%3D%2025%20A.>

<https://www.cynergiebikes.com/about-ebikes->

<s/117.htm#:~:text=Depending%20on%20the%20capacity%20of,you%20%2D80%2B%20miles.>

<https://www.bicycling.com/bikes-gear/a40273900/how-to-safely-charge-your-ebike/>

https://www.electronics-notes.com/articles/basic_concepts/current/electrical-current-units-ampere.php

<https://www.nagwa.com/en/explainers/980189686841/#:~:text=We%20simply%20find%20current%20by,charge%2C%20and%20%F0%9D%91%A1%20represents%20time.>

<https://www.generalrecreationinc.com/solar-charging-stations/>

(20) Capsule Curriculum Vitae(One-page brief C.V. only)